



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP3623

Database Programming

2025/2026 SEMESTER 1

Lab Assignment 2

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Section	02

Part A Data Preparation

Question A1

Class > clinicdb

>_ Open MongoDB shell

+ Create collection

↻ Refresh

Collection name ↕	Properties ↕	Storage size ↕	Documents ↕	Avg. document size ↕	Indexes ↕	Total index size ↕
appointments	-	4.10 kB	0	0 B	1	4.10 kB
loginSessions	-	4.10 kB	0	0 B	1	4.10 kB

Question A2

Insert documents into appointments collection.

```
db.appointments.insertMany([
  {
    appointmentId: 1,
    patientName: "Stella Liong Eh Ha",
    doctor: "Nur Hjh Syaidatul Fahmi binti Jumat",
    department: "Surgical",
    status: "COMPLETED",
    fee: 1500,
    appointmentDate: ISODate("2025-12-01")
  }, {
    appointmentId: 2,
    patientName: "Mohammed Zaidi Mustaqim bin Abidin",
    doctor: "Nur Hjh Syaidatul Fahmi binti Jumat",
    department: "Surgical",
    status: "PENDING",
    fee: 1600,
    appointmentDate: ISODate("2025-12-02")
  }, {
    appointmentId: 3,
    patientName: "Jane Dhillon",
    doctor: "Yeow Jye Khi",
    department: "Otorhinolarygology",
    status: "COMPLETED",
    fee: 136,
    appointmentDate: ISODate("2025-12-03")
  }, {
    appointmentId: 4,
    patientName: "Seow Luk Tiah",
    doctor: "Yeow Jye Khi",
    department: "Otorhinolarygology",
    status: "COMPLETED",
    fee: 60,
    appointmentDate: ISODate("2025-12-04")
  }, {
    appointmentId: 5,
    patientName: "Mohammed Hj Rafizal Nasir bin Fauzul Jamaluddin",
```

```

        doctor: "Gunasekaran Rao",
        department: "Anaesthesiology",
        status: "PENDING",
        fee: 364,
        appointmentDate: ISODate("2025-12-05")
    }, {
        appointmentId: 6,
        patientName: "Haji Zaiyon Hazizan bin Nik Jamil Dahlan",
        doctor: "Ambiga a/l Cecil",
        department: "Paediatric",
        status: "CANCELLED",
        fee: 136,
        appointmentDate: ISODate("2025-12-06")
    }, {
        appointmentId: 7,
        patientName: "Jane Arumugam",
        doctor: "Ambiga a/l Cecil",
        department: "Paediatric",
        status: "COMPLETED",
        fee: 600,
        appointmentDate: ISODate("2025-12-07")
    }, {
        appointmentId: 8,
        patientName: "John Prakash",
        doctor: "Gunasekaran Rao",
        department: "Anaesthesiology",
        status: "COMPLETED",
        fee: 164,
        appointmentDate: ISODate("2025-12-08")
    }
}
])

```

Output:

```

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69687baf06a783c652c6d9a1'),
    '1': ObjectId('69687baf06a783c652c6d9a2'),
    '2': ObjectId('69687baf06a783c652c6d9a3'),
    '3': ObjectId('69687baf06a783c652c6d9a4'),
    '4': ObjectId('69687baf06a783c652c6d9a5'),
    '5': ObjectId('69687baf06a783c652c6d9a6'),
    '6': ObjectId('69687baf06a783c652c6d9a7'),
    '7': ObjectId('69687baf06a783c652c6d9a8')
  }
}

```

Insert documents into loginSessions collection.

```
db.loginSessions.insertMany([
  {
    staffId: 1,
    staffName: "Nur Hjh Syaidatul Fahmi binti Jumat",
    role: "Doctor",
    status: "INACTIVE",
    ipAddress: "197.178.155.216",
    loginTime: ISODate("2025-12-08T18:17:15")
  }, {
    staffId: 2,
    staffName: "Yeow Jye Khi",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "197.178.155.217",
    loginTime: ISODate("2025-12-08T19:48:11")
  }, {
    staffId: 3,
    staffName: "Gunasekaran Rao",
    role: "Doctor",
    status: "INACTIVE",
    ipAddress: "197.178.155.210",
    loginTime: ISODate("2025-12-08T00:10:17")
  }, {
    staffId: 5,
    staffName: "Ambiga a/l Cecil",
    role: "Doctor",
    status: "INACTIVE",
    ipAddress: "197.178.155.110",
    loginTime: ISODate("2025-12-08T08:14:41")
  }, {
    staffId: 6,
    staffName: "Tay Gek Sian",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "197.178.155.116",
    loginTime: ISODate("2025-12-08T06:17:15")
  }, {
    staffId: 7,
    staffName: "Nur Camelia Talib binti Muhaimen Rasyid",
    role: "Nurse",
    status: "INACTIVE",
    ipAddress: "197.178.155.117",
    loginTime: ISODate("2025-12-08T20:48:11")
  }, {
    staffId: 8,
    staffName: "Melinda Siong Tok Rui",
    role: "Nurse",
```

```

        status: "INACTIVE",
        ipAddress: "197.178.155.110",
        loginTime: ISODate("2025-12-08T19:10:17")
    }, {
        staffId: 9,
        staffName: "Noratasha Nafiz binti Che Midali",
        role: "Nurse",
        status: "ACTIVE",
        ipAddress: "197.178.155.130",
        loginTime: ISODate("2025-12-08T05:14:41")
    }
]
])

```

Output:

```

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69687c6206a783c652c6d9a9'),
    '1': ObjectId('69687c6206a783c652c6d9aa'),
    '2': ObjectId('69687c6206a783c652c6d9ab'),
    '3': ObjectId('69687c6206a783c652c6d9ac'),
    '4': ObjectId('69687c6206a783c652c6d9ad'),
    '5': ObjectId('69687c6206a783c652c6d9ae'),
    '6': ObjectId('69687c6206a783c652c6d9af'),
    '7': ObjectId('69687c6206a783c652c6d9b0')
  }
}

```

Part B Appointment Reporting

Question B1 Advanced Query & Sorting

```
db.appointments
  .find(
    {status: "COMPLETED", fee: {$gt: 100, $lt: 300}},
    {appointmentId: 1, patientName: 1, department: 1, fee: 1})
  .sort({department: 1, fee: -1})
```

Output:

```
> db.appointments
  .find(
    {status: "COMPLETED", fee: {$gt: 100, $lt: 300}},
    {appointmentId: 1, patientName: 1, department: 1, fee: 1})
  .sort({department: 1, fee: -1})
< {
  _id: ObjectId('69687baf06a783c652c6d9a8'),
  appointmentId: 8,
  patientName: 'John Prakash',
  department: 'Anaesthesiology',
  fee: 164
}
{
  _id: ObjectId('69687baf06a783c652c6d9a3'),
  appointmentId: 3,
  patientName: 'Jane Dhillon',
  department: 'Otorhinolarygology',
  fee: 136
}
```

Question B2 Aggregation Summary

```
db.appointments.aggregate([
  {$group: {
    _id: "$department",
    totalAppointments: {$sum: 1},
    totalFees: {$sum: "$fee"}},
  {$sort: {totalFees: -1}}
])
```

Output:

```
> db.appointments.aggregate([
  {$group: {
    _id: "$department",
    totalAppointments: {$sum: 1},
    totalFees: {$sum: "$fee"}},
  {$sort: {totalFees: -1}}
])
< {
  _id: 'Surgical',
  totalAppointments: 2,
  totalFees: 3100
}
{
  _id: 'Paediatric',
  totalAppointments: 2,
  totalFees: 736
}
{
  _id: 'Anaesthesiology',
  totalAppointments: 2,
  totalFees: 528
}
{
  _id: 'Otorhinolarygology',
  totalAppointments: 2,
  totalFees: 196
}
```

Part C Clinic Security Monitoring

Question C1 Active Staff Sessions

```
db.loginSessions
  .find(
    {status: "ACTIVE"},
    {staffName: 1, role: 1, ipAddress: 1, loginTime: 1})
  .sort({loginTime: -1})
```

Output:

```
> db.loginSessions
  .find(
    {status: "ACTIVE"},
    {staffName: 1, role: 1, ipAddress: 1, loginTime: 1})
  .sort({loginTime: -1})
< {
  _id: ObjectId('69687c6206a783c652c6d9aa'),
  staffName: 'Yeow Jye Khi',
  role: 'Doctor',
  ipAddress: '197.178.155.217',
  loginTime: 2025-12-08T19:48:11.000Z
}
{
  _id: ObjectId('69687c6206a783c652c6d9ad'),
  staffName: 'Tay Gek Sian',
  role: 'Nurse',
  ipAddress: '197.178.155.116',
  loginTime: 2025-12-08T06:17:15.000Z
}
{
  _id: ObjectId('69687c6206a783c652c6d9b0'),
  staffName: 'Noratasha Nafiz binti Che Midali',
  role: 'Nurse',
  ipAddress: '197.178.155.130',
  loginTime: 2025-12-08T05:14:41.000Z
}
```

Question C2 Distinct & Count

- (a) List all distinct IP addresses used by staff with ACTIVE sessions.

```
db.loginSessions.distinct("ipAddress", {status: "ACTIVE"})
```

Output:

```
> db.loginSessions.distinct("ipAddress", {status: "ACTIVE"})
< [ '197.178.155.116', '197.178.155.130', '197.178.155.217' ]
```


- (b) Calculate the number of unique staff members who currently have ACTIVE sessions.

```
db.loginSessions.distinct("staffId", {status: "ACTIVE"}).length
```

Output:

```
> db.loginSessions.distinct("staffId", {status: "ACTIVE"}).length
< 3
```

Part D Indexing & Data Maintenance

Question D1 Compound Index

```
db.appointments.createIndex({department: 1, status: 1})
```

Output:

```
> db.appointments.createIndex({department: 1, status: 1})
< department_1_status_1
```

Question D2 Update & Delete Operations

- (a) Update all appointments with status CANCELLED to ARCHIVED.

```
db.appointments.updateMany(
  {status: "CANCELLED"},
  {$set: {status: "ARCHIVED"}})
```

Output:

```
> db.appointments.updateMany(
  {status: "CANCELLED"},
  {$set: {status: "ARCHIVED"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- (b) Delete all login session records with status INACTIVE.

```
db.loginSessions.deleteMany({status: "INACTIVE"})
```

Output:

```
> db.loginSessions.deleteMany({status: "INACTIVE"})
< {
  acknowledged: true,
  deletedCount: 5
}
```